



BITS Pilani
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Computer Vision
Computer Vision (S2-25_AIMLZG525)

Session #2: Digital Image Fundamentals Raghavendra Singh

Acknowledgement: Slide Materials adopted from - Intro to Computer Vision (Cornell Tech); Noah Snavely

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Topics

- Image Representation
- Image Digitization (Sampling & Quantization)
- Digital Image Properties

- OpenCV/Python Demonstration

Image representation and image analysis tasks, Image digitization - Sampling and resolution, Digital Images, Digital Image types -Binary, Gray-scale and Color, Color Images, Color spaces: RGB and HSV



Simple Image Formation Model

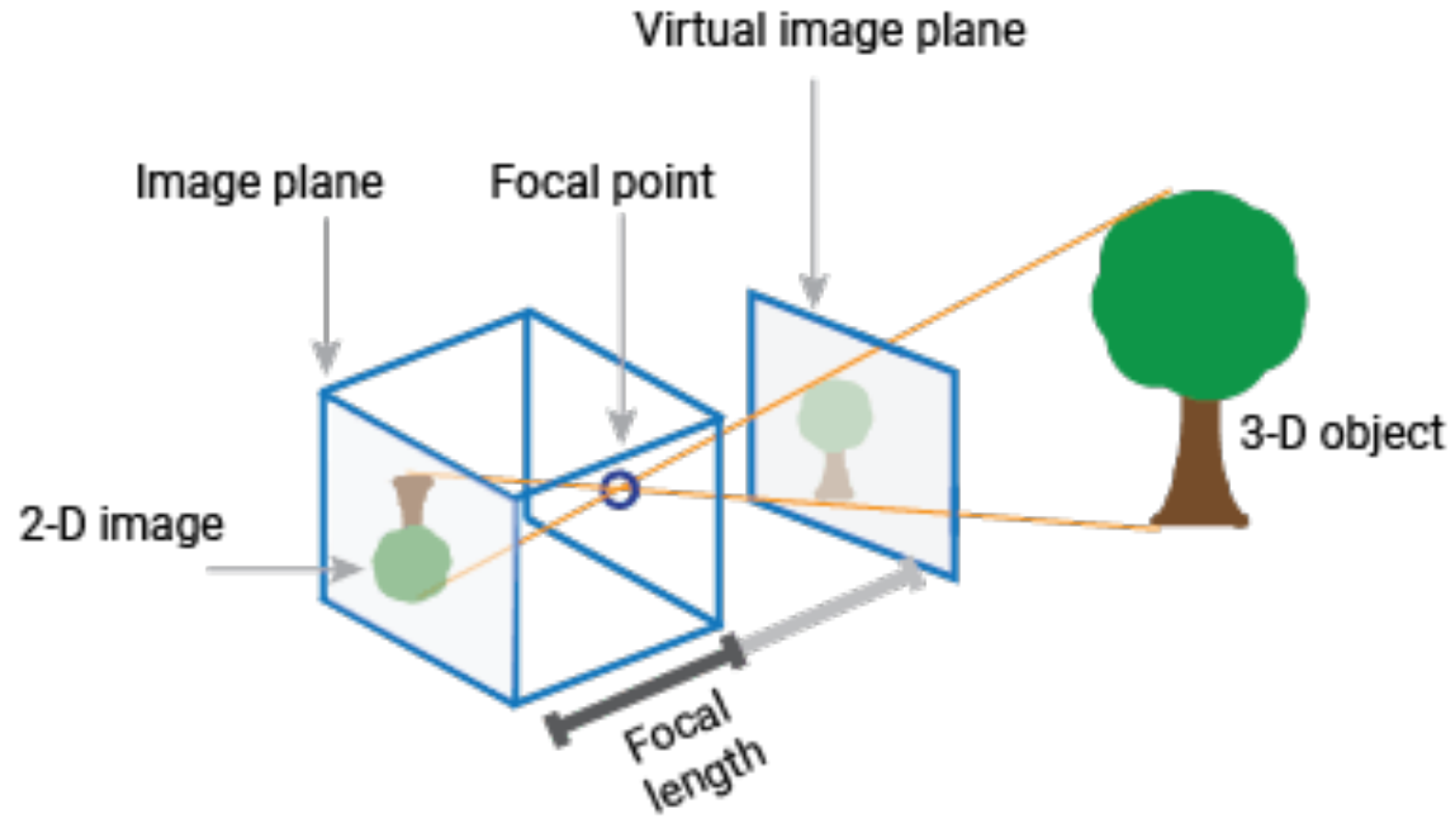
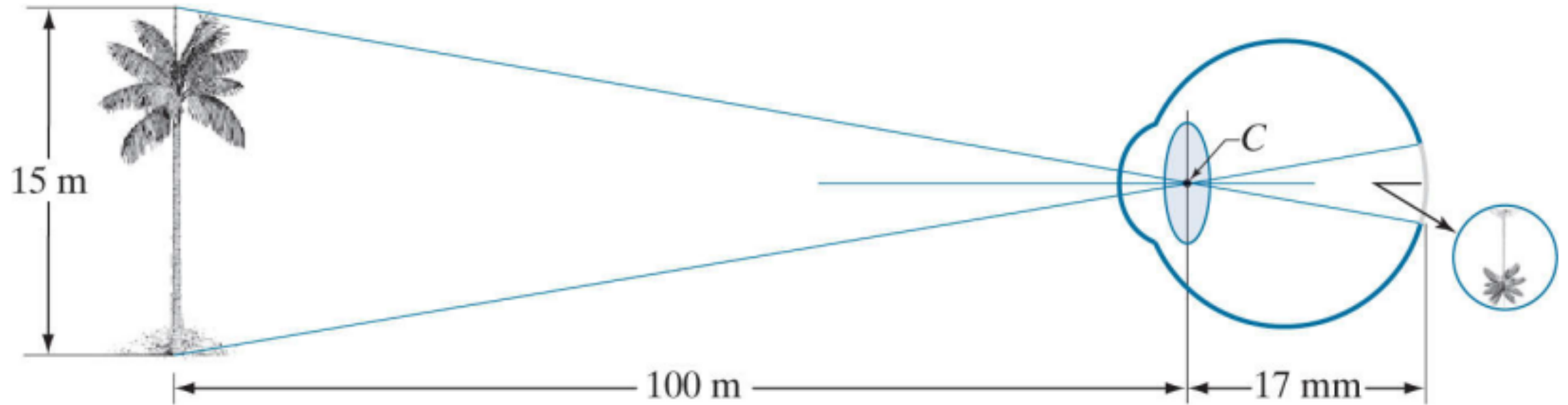
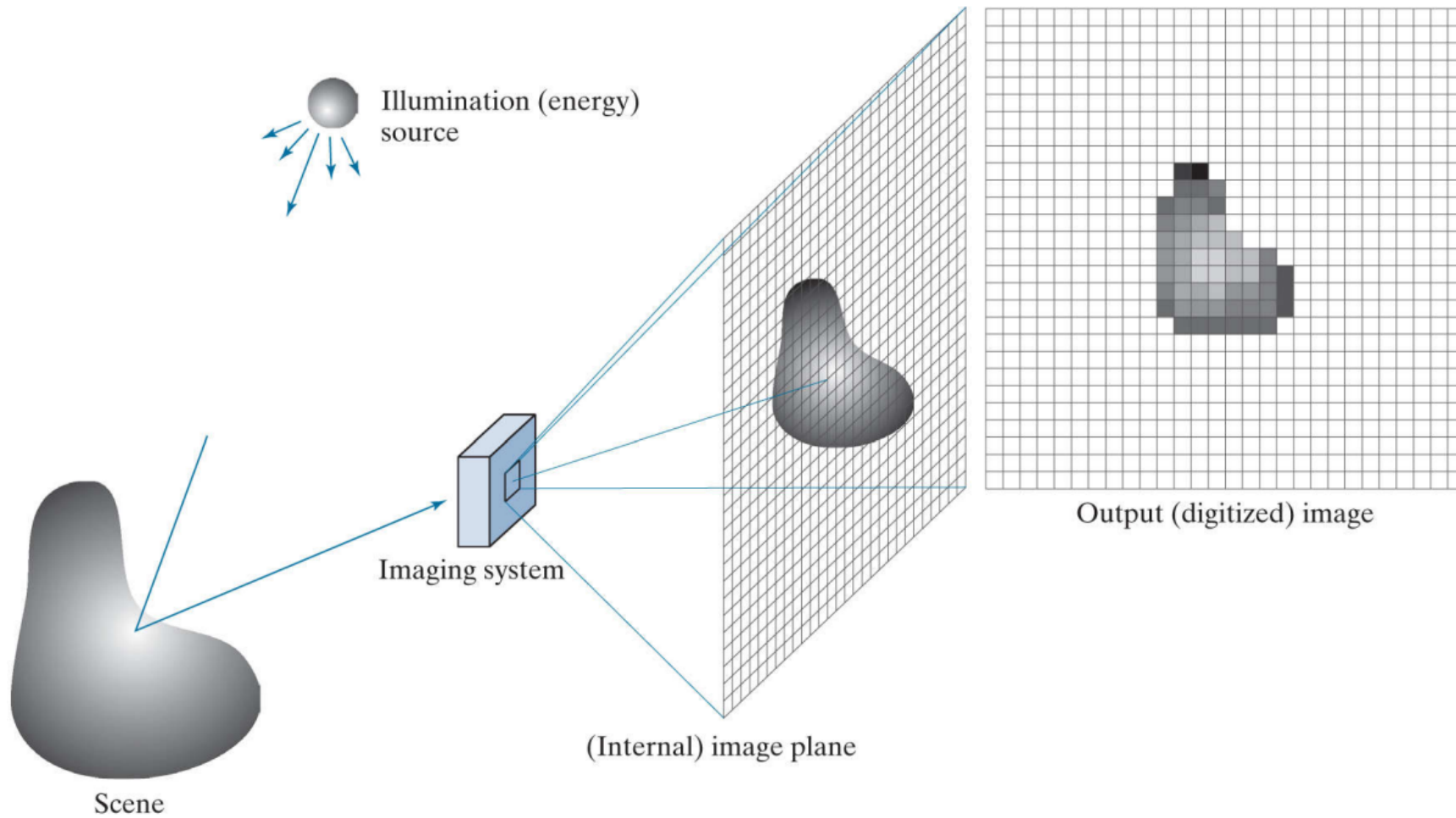


Image Formation in the Eye





Digital Image Acquisition





Simple Image Formation Model

- We denote images by two-dimensional functions of the form $f(x, y)$.
 - Value of f at spatial coordinates (x, y) is a scalar quantity;
 - Physical meaning of $f(x, y)$ is determined by the source of the image, and whose values are proportional to energy radiated by a physical source (e.g., electromagnetic waves);

$$0 \leq f(x, y) < \infty$$

- Function $f(x, y)$ is characterized by two components:
 - Amount of source illumination incident on the scene being viewed, and
 - Amount of illumination reflected by the objects in the scene.

$$f(x, y) = i(x, y)r(x, y)$$

$$0 \leq i(x, y) < \infty$$

$$0 \leq r(x, y) \leq 1$$



Simple Image Formation Model - Grayscale Image

- Let the intensity (gray level) of a monochrome image at any coordinates (x, y) be denoted by

$$\ell = f(x, y) \quad L_{\min} \leq \ell \leq L_{\max}$$

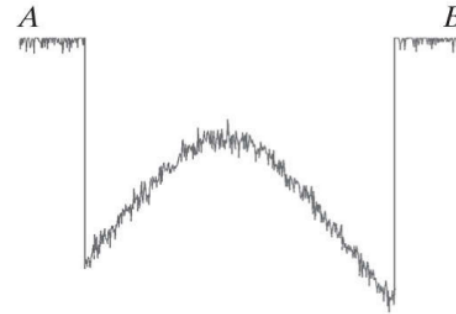
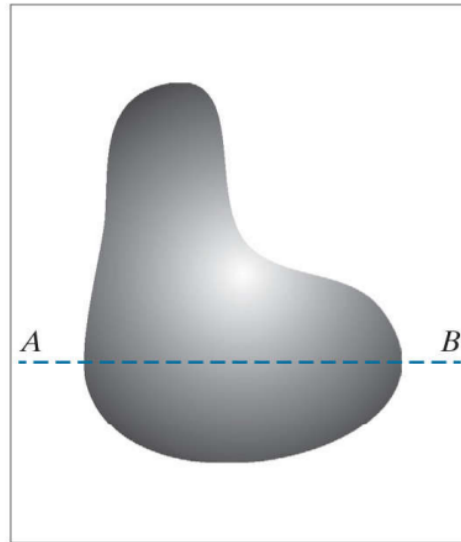
$$L_{\min} = i_{\min} r_{\min} \quad L_{\max} = i_{\max} r_{\max}$$

- For a gray scale, the range is scaled between $[0, L-1]$;
 - All intermediate values from 0 to $L-1$ are shades of gray varying from black to white.



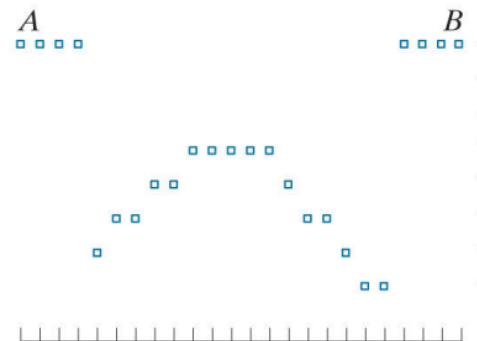
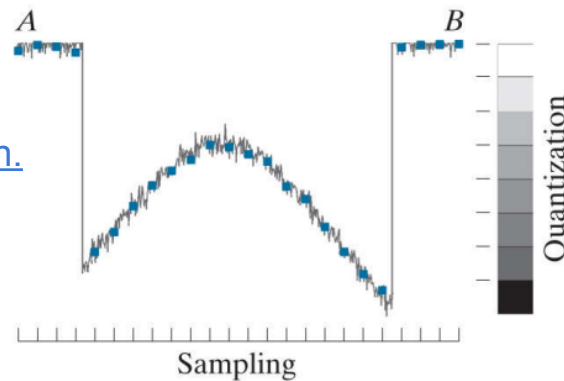
Sampling and Quantization

Continuous image



A scan line showing intensity variations along line AB in the continuous image

Sampling and quantization.



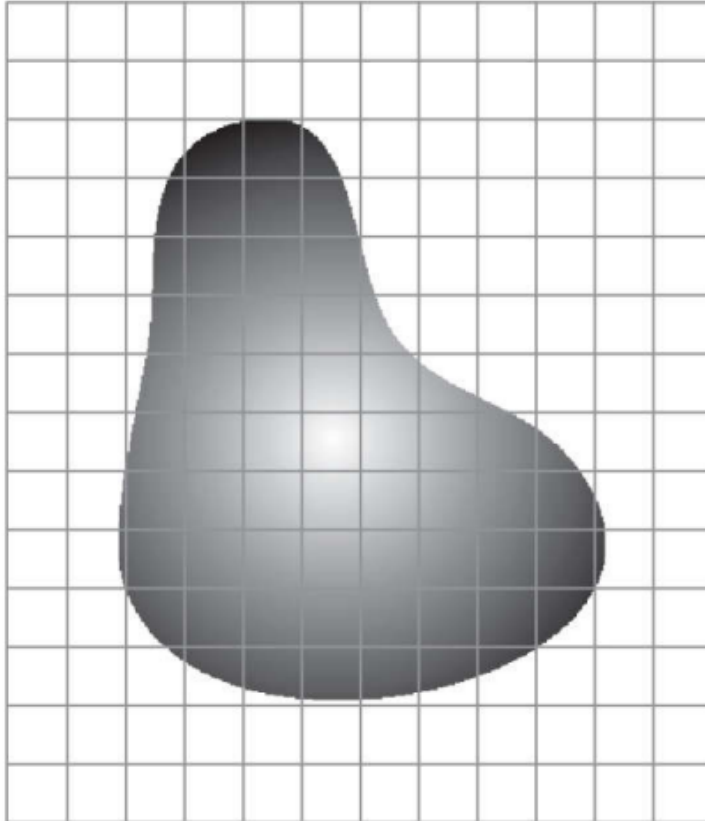
Digital scan line

Digitizing the coordinate values is called *sampling* ; Digitizing the amplitude values is called *quantization*.

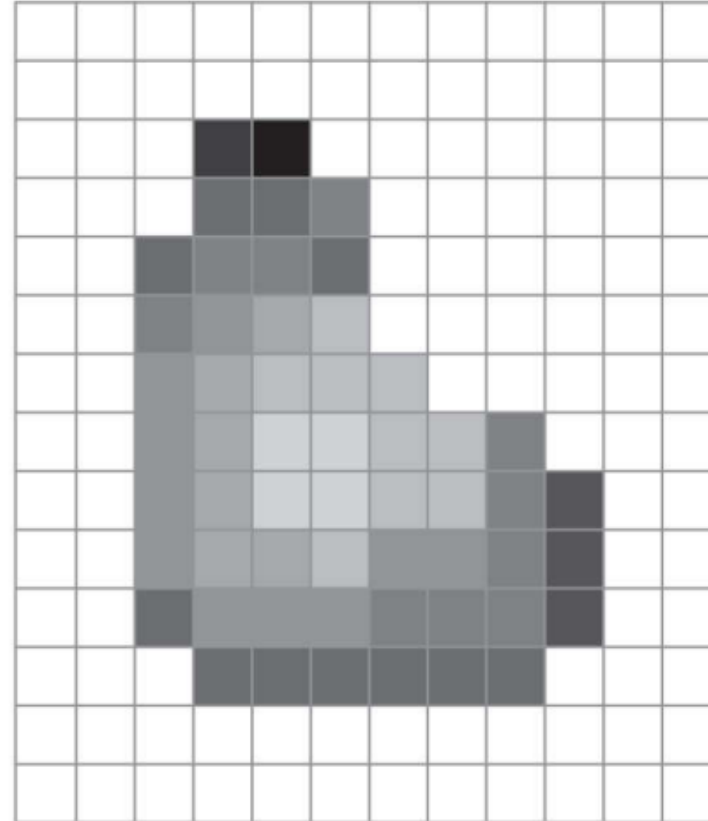


Sampling and Quantization

Continuous image projected onto a sensor array



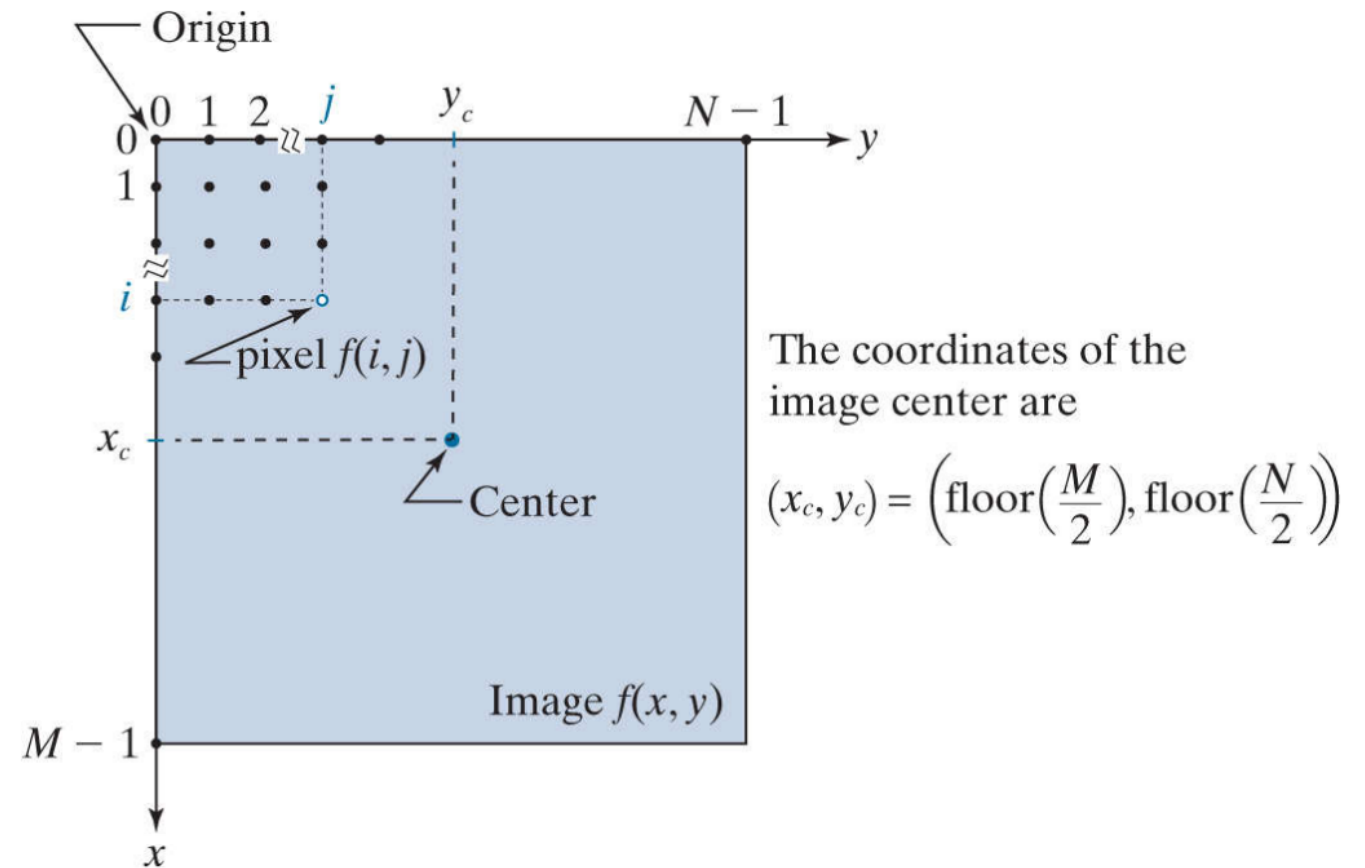
Result of image sampling and quantization





Matrix Representation of an Image

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(0, 1) & \cdots & f(0, N-1) \\ f(1, 0) & f(1, 1) & \cdots & f(1, N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1, 0) & f(M-1, 1) & \cdots & f(M-1, N-1) \end{bmatrix}$$



Spatial & Intensity Resolution

- **Spatial resolution** is a measure of the smallest discernible detail in an image;
- **Intensity resolution** similarly refers to the smallest discernible change in intensity level

930 dpi



300 dpi



150 dpi



72 dpi





Spatial & Intensity Resolution

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256-level



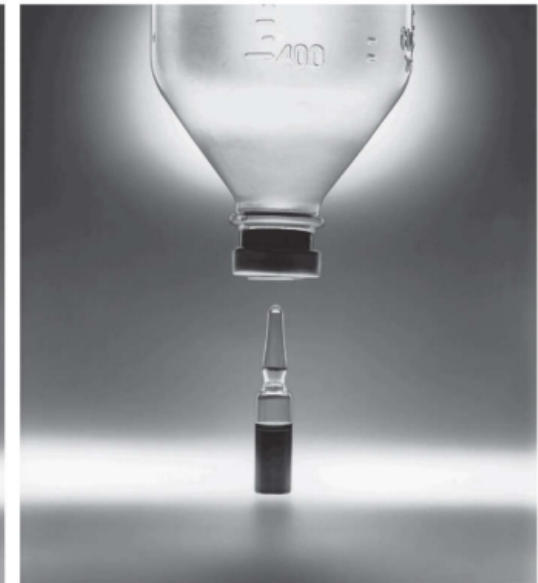
128-level



64-level



32-level



All images have spatial resolution 2022 × 1800

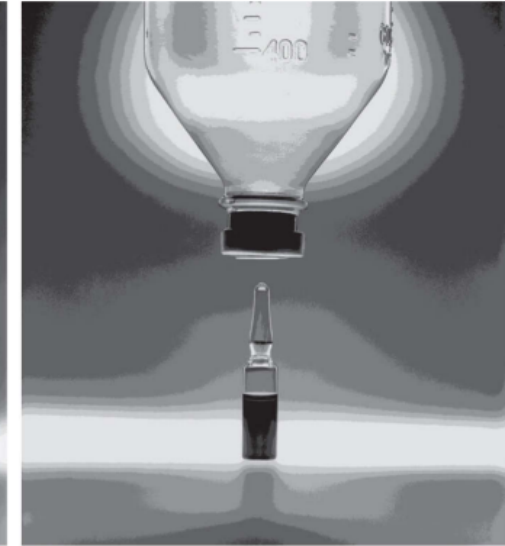
Spatial & Intensity Resolution

- **Spatial resolution** is a measure of the smallest discernible detail in an image;
- **Intensity resolution** similarly refers to the smallest discernible change in intensity level

16-level



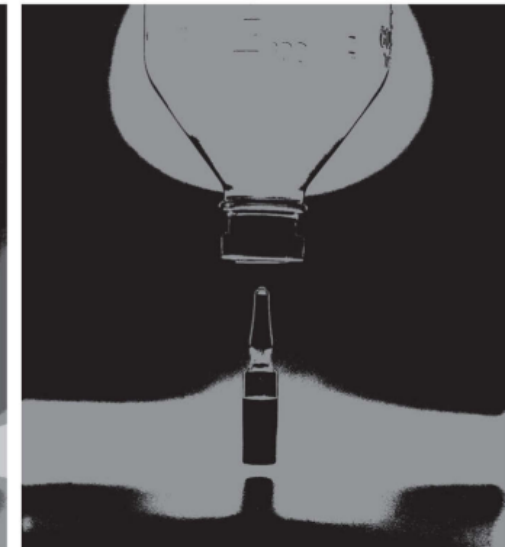
8-level



4-level



2-level



All images have spatial resolution 2022 × 1800



- Readings:

Digital Image Processing, Rafael C. Gonzalez & Richard E woods, Third Ed, Chapter 2



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Thank you